AI GIS Road and Sidewalk Assessment Platform

Project requirements and UI UX design specification

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What we are building

A web platform that turns geotagged road and sidewalk photos into a map of suspected defects. AI suggests what it sees; a qualified reviewer confirms or corrects each finding. Verified findings help municipal and engineering teams plan inspections, prioritize maintenance, and export condition records.

How someone will use it

A project manager creates a survey project. A field collector uploads photos with locations. The system processes the images and creates candidate findings. A reviewer checks the photos and map pins. A manager reviews the verified findings, selects inspection priorities, and generates a report.

Recommended first release

Build a road-focused pilot with photo upload, map viewing, AI suggestions, human review, a configurable priority score, and exports. Include a manual sidewalk finding type from the start; enable automated sidewalk classification only after a dedicated dataset and validation are available.

Outcome	What the product delivers
See conditions	A shared map with source photos, finding types, review status, and location quality.
Trust the record	Human-reviewed results with an audit history and preserved original AI output.
Plan field work	A sorted inspection list using verified severity and configurable context.
Share results	Project reports, CSV, and geographic data exports for existing workflows.

Document scope

The following pages define users, workflow, release features, screen behavior, illustrative UI layouts, development architecture, data requirements, delivery stages, and acceptance criteria. Priorities and schedule are proposed planning assumptions; validation targets must be agreed with the pilot partner.

Product boundaries

The platform supports condition screening and inspection planning. It does not automatically certify accessibility compliance, structural safety, exact defect dimensions, or engineering repair specifications.

1 Users and access

User	Main tasks	Access
Organization administrator	Invite staff, manage roles, configure organization settings.	All projects within their organization.
Project manager	Create projects, select survey area, monitor coverage, configure priorities, export reports.	Assigned projects; project management permissions.
Field collector	Capture or upload photos, add notes, confirm location, retry failed uploads.	Assigned project uploads and own submission status.
Technical reviewer	Verify, edit, reject, or merge candidate findings; confirm severity and location.	Assigned review queues and finding history.
Viewer	Explore verified findings and approved reports.	Read-only access to assigned projects.

Access requirements

Every record belongs to an organization and project. The server must check access on every request, including image downloads, job status, and exports. Users cannot access another organization by changing a URL or record ID. Invitations expire, removed users lose access, and role changes are recorded.

Typical use case

A municipal team surveys a selected road corridor. Collectors submit street-level photos. The AI marks possible potholes and surface cracking. A reviewer rejects shadows and confirms actual defects. The manager views only verified findings, prioritizes field checks, and exports a list with photos and geographic coordinates.

Project setup fields

Required: project name, organization, project manager, survey area, supported finding types, and assigned reviewers. Optional: survey dates, road category, public-facility context, inspection history, and partner reference number.

Release assumptions

V1 uses a responsive desktop workspace for managers and reviewers, plus a mobile web upload flow for collectors. Authorized images and a qualified reviewer are supplied by the pilot partner. A native mobile application, automated video ingestion, and offline capture are later features.

2 How the platform works

Step	User action	System behavior
1 Create project	Select area, finding types, team, and survey dates.	Save boundary and configuration; show an empty project with an upload action.
2 Submit photos	Choose images and confirm consent and location.	Validate files, record metadata, flag missing GPS, and show per-file upload progress.
3 Analyze	Start analysis for eligible images.	Queue background jobs; show queued, processing, completed, or failed status.
4 Generate findings	Open results when processing completes.	Store candidate type, confidence, image overlay, source location, and model version.
5 Human review	Inspect photo, type, location, and context.	Save verified, rejected, or needs-field-check status; preserve edits and reviewer identity.
6 Prioritize	Choose verified findings and priority settings.	Calculate a transparent inspection priority and explain each contributing factor.
7 Share and revisit	Export report or return after field inspection.	Create a dated snapshot; allow inspection notes and resolution evidence.

Finding lifecycle

AI generated → Awaiting review → Verified, Rejected, or Needs field check. A verified finding may progress to Inspection planned → Inspected → Resolved. “Resolved” requires a note or evidence photo and an authorized user. Reopening a record retains its full history.

Exceptions users must understand

No detection means no candidate was produced; it does not prove the road is defect-free. A missing or unreliable location requires confirmation before the finding is included in a geographic export. Blurry or unsupported images are flagged with a reason and a replacement action.

Location rule

A photo GPS point is an approximate observation location, not automatically the exact defect coordinate. Reviewers can move a pin; the system retains the original coordinate, corrected coordinate, and location quality. Exact image-to-ground positioning is a separate validated capability.

3 Feature list and release priorities

Feature	V1 requirement	Later release
Projects and team	Organization roles, project boundaries, assignments, status.	Cross-project portfolio planning.
Photo collection	Batch JPEG/PNG upload, metadata extraction, location confirmation, retry.	Offline capture, guided routes, video ingestion.
Quality screening	Flag blur, missing GPS, unsupported files, and exact duplicate images.	Coverage planning and richer capture guidance.
AI road screening	Proposed classes: potholes, surface cracking, pavement deterioration.	Expanded classes after measured validation.
Sidewalk observations	Manual photo-backed observations with human review.	AI condition screening after dataset validation.
Map and list	Synchronized pins and list; filter by type, severity, status, and date.	Temporal comparison and route optimization.
Review workspace	Overlay toggle, editable type, severity, pin, notes, verify/reject/field-check.	Review assignment rules and second-review sampling.
Prioritization	Configurable contextual scoring with explanation; missing-data flag.	Budget scenarios and validated cost estimates.
Reports and exports	CSV, GeoJSON, and a PDF summary of verified records.	Direct integrations with partner asset systems.
Inspection follow-up	Assigned owner, planned date, outcome notes, resolution photo.	Work orders, notifications, maintenance scheduling.
Audit and privacy	Review history, protected media, retention settings, export logging.	Advanced governance and enterprise access controls.

Priority score behavior

Use configurable weights for verified severity, road context, and proximity to selected public facilities. Confidence is a review aid, not a severity score. Show inputs, weights, missing values, and the resulting score. Label the score “inspection priority”; never present unverified candidates as confirmed maintenance needs.

Suggested project dashboard

Show uploaded images, processing progress, findings awaiting review, verified findings, and records needing location confirmation. Add reviewed findings by type and a coverage map. Avoid a “percentage surveyed” metric unless the expected route or asset inventory has been defined.

4 UI UX screens and behavior

Screen	Layout and primary action	Important behavior
Project overview	Project list, survey status, upload and review counts; Create project.	Empty state explains the first step. Show only authorized projects.
New project	Short form with boundary map, team, dates, and finding types; Save project.	Inline validation; preserve entered fields when a save fails.
Upload photos	Drop zone, file queue, GPS indicator, privacy confirmation; Upload.	Per-file progress and retry; successful files are not re-uploaded.
Map and findings	Filters above map; findings list and selected record drawer; Open review.	Map and list selection stay synchronized. Rejected records are hidden by default.
Review finding	Large photo and overlay; type, severity, location, notes; Verify.	Require explicit type and severity for verification; warn about unconfirmed location.
Priority and inspection	Verified list with score breakdown, owner and date; Plan inspection.	Exclude rejected candidates; flag incomplete context instead of silently inventing inputs.
Reports	Project, date, status and geographic filters; Generate report.	Default to verified results; label optional unreviewed data prominently.
Settings	Team roles, finding categories, priority settings, retention; Save changes.	Confirm consequential changes and record who changed them.

Visual design

Use a calm, professional interface with a persistent left navigation, clear page titles, generous spacing, and one prominent primary action per screen. Use restrained color for status and severity; pair colors with labels and icons. Keep AI suggestions visually distinct from reviewed findings.

Accessibility and responsive behavior

Provide keyboard navigation, visible focus states, readable contrast, labeled controls, and zoomable photos. Offer the findings list as an alternative to the map. On mobile, use a single-column upload flow, large touch controls, camera/file selection, and clear connectivity feedback.

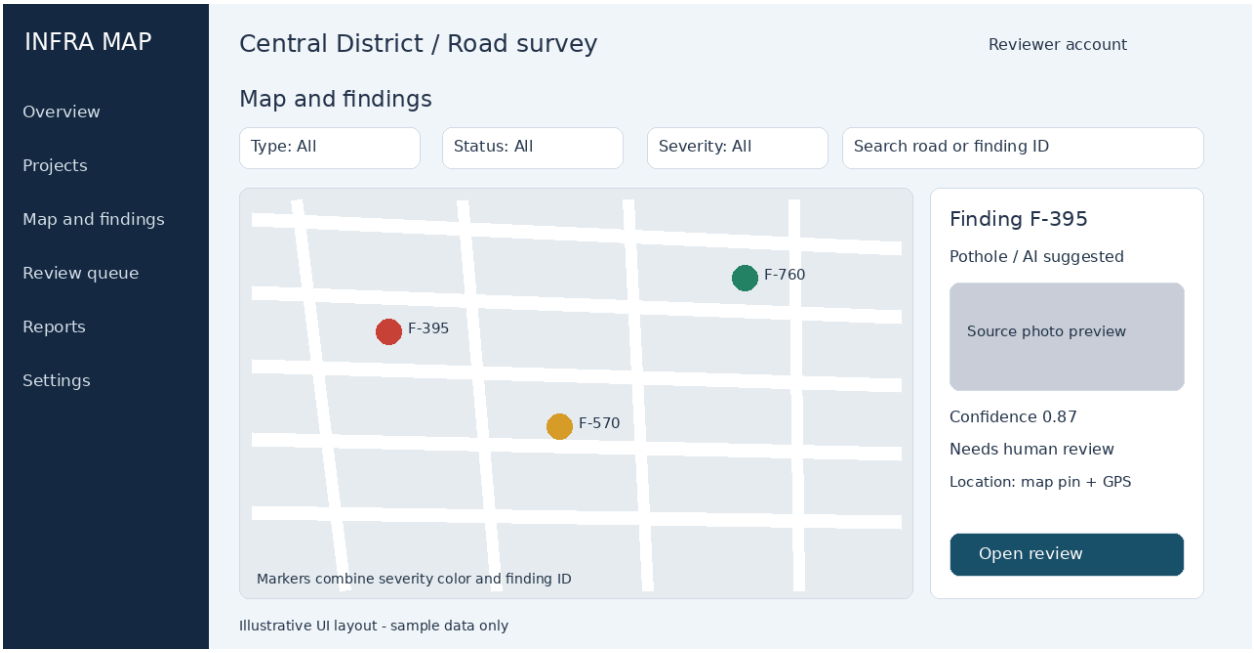
Loading and failure states

Show background processing progress without blocking navigation. Explain failures in plain language with retry or replace-photo actions. Empty filters show “No findings match these filters” and a reset action. Unsaved review edits trigger a navigation warning.

5 Illustrative screen layouts

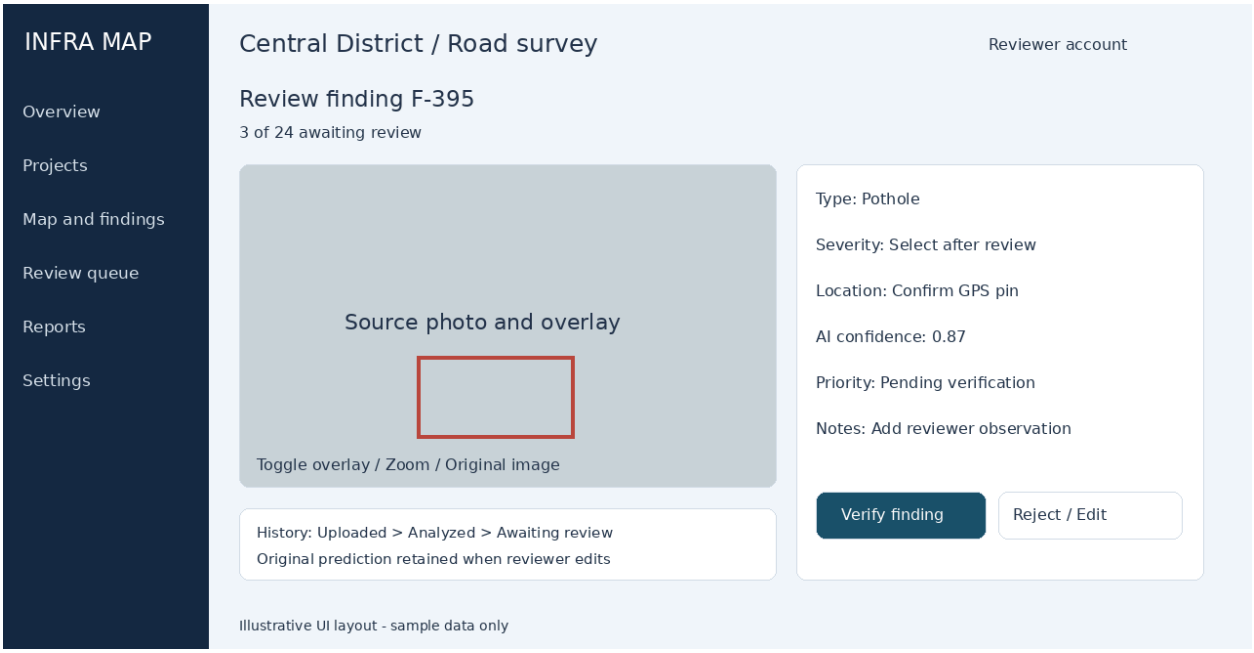
These wireframes show the proposed arrangement and interaction hierarchy. Labels and values are sample data, not project results.

Map and findings



Select a pin or list record to open the same finding drawer. The photo, review status, and location quality stay visible before the user opens the full review workspace.

Finding review



Reviewers inspect the original photo and optional AI overlay, correct the record, and choose Verify, Reject, or Needs field check. The next record opens only after the current decision has saved.

6 Development architecture and data

Proposed architecture

Build a responsive web client connected to an authenticated application API. Store project and geographic records in a relational database with spatial capabilities. Keep photos in private object storage. Background workers perform quality checks and AI inference. The web client polls job status in V1; report workers generate downloadable exports.

Component	Responsibility
Web application	Project setup, upload, map/list, review, prioritization, and reports.
Application API	Authentication, access checks, metadata, reviews, inspection updates, exports.
Spatial database	Organization-scoped records, project boundaries, findings, coordinates, audit events.
Private media storage	Original images, derived previews and overlays; time-limited authorized download links.
Job queue and workers	Upload checks, inference, retries, progress events, and report generation.
AI service	Versioned detection output; predicted type, confidence, image region, and failure reasons.
Monitoring	Job errors, API failures, queue age, processing duration, storage use, and audit logs.

Core data entities

Organization and UserMembership define tenancy and roles. Project holds area and configuration. Image holds source file, capture time, GPS, quality flags and uploader. AnalysisJob tracks state and model version. Finding stores predicted type, original and corrected location, review status and linked image. ReviewEvent preserves each decision. Inspection stores owner, planned date and outcome. Report stores export filters, timestamp and file location.

Minimum API operations

Create/list projects; invite/manage project members; request upload and confirm completion; start/retry analysis; retrieve jobs; query findings by project and map extent; submit review; update inspection; request export and retrieve export status.

Data integrity rules

Require a project link for every image and finding. Preserve original predictions separately from human edits. Make upload completion and job creation idempotent. Reject conflicting reviews with a reload prompt. Deleting or archiving data follows configured retention and keeps a permitted audit record.

7 AI development and validation

Start with a measured baseline

Obtain authorized, representative street-level images for the chosen road classes. Define label examples and ambiguous cases with a qualified reviewer. Create geographic and time-separated training, validation, and test sets; near-duplicate photos from the same road segment must not leak across sets.

Model output contract

For each candidate, return a class, confidence, image region, model version, and linked source image. Attach the image location as an approximate observation point. Do not infer exact depth, dimensions, repair cost, or regulatory compliance from an ordinary photo.

Validation plan

Measure	How to evaluate	Release use
Precision and recall	Compare predictions with reviewer-labeled held-out images, per class.	Agree thresholds with pilot partner before training or tuning.
Reviewer agreement	Have independent reviewers label a representative subset.	Resolve ambiguous labels and update annotation guidance.
Condition robustness	Break results down by lighting, surface, blur, geography and capture device.	Document weak conditions and trigger review or resubmission.
Location quality	Compare original/corrected pins with verified observation locations.	Define acceptable accuracy for the specific partner workflow.
Screening time	Compare a defined task with and without AI assistance.	Report sample size and actual time saved, including review.
Workflow usability	Observe collectors, reviewers and managers completing key tasks.	Fix blocked or misunderstood steps before pilot expansion.

Confidence and severity

Choose confidence thresholds using validation results, not an arbitrary default. All AI candidates require human review in V1. Severity follows a documented reviewer rubric; dimensions or severity that cannot be established from the image become “Needs field check.”

Pilot deliverables

Provide a working prototype, dataset and labeling guide, model evaluation report, reviewed geographic export, user feedback summary, and limitations register. Demonstrate the entire upload-to-report workflow using real authorized data.

Privacy

Restrict media access and define retention with the partner. Identify faces and license plates for masking before externally shared imagery when required by the agreed policy. Never use partner photos for model training without the necessary rights and explicit project authorization.

8 How to develop and deliver

Suggested planning sequence: eight two-week sprints, subject to dataset availability, reviewer access, and integration needs. Workstreams can overlap. A road pilot is the first delivery target; automated sidewalk screening has a separate readiness gate.

Sprint	Build focus	Reviewable output
1	Confirm pilot scope, roles, data permissions, label guide and screen designs.	Approved feature scope, clickable design, and acceptance checklist.
2	Organization access, project setup, database and private media storage.	Users can create authorized projects and upload sample files.
3	Batch upload, GPS confirmation, quality checks and job queue.	Reliable uploads with progress, error reasons, and retry.
4	AI baseline and versioned inference integration.	Candidate findings from a held-out sample with recorded evaluation.
5	Map/list synchronization, filters, finding drawer and full review.	Reviewers can verify, reject, edit, and inspect history.
6	Priority rules, inspection tracking, CSV/GeoJSON/PDF exports.	A verified finding can reach inspection planning and report export.
7	Security checks, usability sessions, performance measurements and pilot run.	Pilot feedback, defect fixes, measured model and workflow results.
8	Deployment readiness, documentation, training and final handover.	Operational pilot with monitoring, backup/restore check and user guide.

Team responsibilities

Product lead owns scope and partner decisions. UI/UX designer owns flows and screen specifications. Frontend engineer owns responsive interactions and map behavior. Backend engineer owns tenancy, data, uploads, jobs and exports. ML/geospatial engineer owns datasets, model evaluation and location handling. QA owns workflow, security and failure-path testing. A qualified infrastructure reviewer defines labels and validates outcomes.

Development approach

Deliver vertical slices: first upload to stored photo, then analysis to candidate, then candidate to verified finding, then verified finding to report. Use separate development, staging, and production environments. Keep test data isolated, configuration versioned, and production secrets outside source control.

Inputs needed before kickoff

Confirm the first partner, survey area, photo source, class definitions, reviewer, location accuracy needs, retention rules, expected volume, and report format. Estimate cost only after these inputs and the selected deployment architecture are agreed.

9 Acceptance criteria and handover

Area	Acceptance condition
Organization isolation	A user cannot retrieve another organization’s projects, images, findings, jobs or reports, including by altering IDs.
Upload reliability	Each file shows success or an actionable failure. Retrying failed uploads does not duplicate successful images or analysis jobs.
Analysis traceability	Each result identifies its source image, model version and processing state. Failed jobs can be retried without duplicate findings.
Review integrity	Verification records user, time, type, severity and decision. Original predictions remain available; concurrent edits cannot silently overwrite decisions.
Map behavior	Filters and selection match between map and list. Missing GPS records remain accessible and are flagged for correction.
Priority rules	Only eligible reviewed findings are scored. Users can inspect weights, inputs and missing context; configuration changes are recorded.
Reporting	Export counts match chosen filters. Reports identify generation date, review status and geographic limitations; fields open correctly in the agreed partner tools.
Inspection updates	Authorized users can plan, complete and reopen inspections; resolution evidence and history remain linked.
Model readiness	Per-class results meet thresholds agreed in advance on held-out pilot data, with limitations and sample counts documented.
Usability and accessibility	Representative users complete upload, review and export tasks; keyboard and non-map list flows work.
Operations	Expected pilot volume and response targets are measured before rollout. Monitoring, backup restore, retention and support ownership are verified.

Handover package

Deliver approved requirements and designs, source code, deployment instructions, configuration reference, dataset and model documentation, API specification, QA results, pilot evaluation, admin guide, user walkthrough, and support procedures.

Future expansion

After the road pilot meets its acceptance gate, prioritize automated sidewalk observations, repeat-survey comparisons, native or offline collection, and direct asset-system integration using partner demand and validation evidence.

Release decision

The product lead and pilot partner confirm workflow readiness, qualified reviewers confirm assessment limitations, engineering confirms operational readiness, and QA confirms the agreed acceptance checks. Unresolved limitations are visible in the product and handover notes.

10 How AI and GIS work together

AI answers “What might be wrong in this photo?” GIS answers “Where was it observed, which asset does it relate to, and what surrounds it?” A human reviewer confirms the condition and location before the record drives inspection planning. The workflow below is the proposed implementation, not a claim of existing model performance.

Step 1 Collect images and location

A collector uploads an authorized street-level photo with its capture time and available GPS metadata. Mobile collection can record device coordinates when the user grants location access. Keep the original image, original coordinates, capture method and any available accuracy estimate. If GPS is missing, ask the user to select a point on the map. Location source is always visible: photo GPS, device GPS, or manually selected.

Step 2 Check image quality

The upload worker checks file type, size, duplicate file hash and required metadata. A quality-screening stage flags blur, darkness or a view that does not show a usable road surface. Each rejected or questionable image has a reason and a replacement action. Quality rules are tuned on pilot imagery; they cannot guarantee a defect is visible.

Step 3 AI detects candidate defects

Use a computer-vision detection model trained or fine-tuned on labeled road images. Start with potholes, surface cracking and pavement deterioration. A detector returns the suspected class, confidence score and bounding box in image pixels. A segmentation model may later produce a more precise image mask. For example, a candidate can be “Pothole, model confidence 0.87, image region [x1, y1, x2, y2].” The confidence is a model score; it is not automatically an 87% real-world correctness probability.

This does not require a chatbot to decide what a defect is. The core screening component is the image model. A language model may later help draft summaries from verified records, but it must not invent defects, coordinates, severity or repair recommendations.

Step 4 GIS places the observation

Create a candidate finding linked to the image and its geographic observation point. Keep image pixel coordinates separate from latitude and longitude. Ordinary photo GPS normally identifies where the camera was, not the exact point of the pothole. Show the pin as approximate until a reviewer confirms or corrects it. Preserve the original point and every correction. A future precise projection workflow would require calibrated camera geometry plus reliable pose and ground/depth information; it is outside V1.

Step 5 GIS adds asset context

Import authorized road centerlines, project boundaries, asset identifiers and relevant facility layers. Spatial queries identify which project area contains the observation, nearby road segments and facilities within configured distances. Suggest a road association rather than silently attaching the nearest road: parallel roads, intersections and weak GPS can create errors. The reviewer confirms ambiguous associations.

Store locations in a spatial database. For example, PostgreSQL with PostGIS supports geographic distance queries such as ST_DWithin using geography distances in meters. Use consistent coordinate reference systems, spatial indexes and an appropriate projected system when a task requires local metric geometry. Preserve source, update date and licensing information for each imported geographic layer.

Step 6 Resolve possible duplicates

Different images may show the same defect. Suggest possible duplicates using compatible type, confirmed road segment, capture time, spatial proximity and image evidence. GPS proximity alone is insufficient: two adjacent potholes must remain separate. Let a reviewer merge candidates into one finding with multiple source images and a preserved history.

Step 7 Human review and inspection priority

The reviewer sees the image, AI overlay, model score, map pin and geographic context. They confirm or change the class, assign severity using a documented rubric and confirm the location. If an image cannot establish a condition, choose Needs field check. Only eligible verified records enter the inspection priority list.

A proposed score can combine reviewed severity, road category and proximity to selected facilities. Use partner-approved weights and show the contribution of each input. Missing context is flagged instead of assumed. AI confidence must not stand in for defect severity. The score ranks inspection attention; it does not decide repair specifications or compliance.

Step 8 Deliver the record

The finished record contains a finding ID, project and asset link, original image, prediction and model version, corrected location, location quality, reviewed type and severity, reviewer decision, contextual inputs and audit history. Managers filter the map, assign a field inspection and export reviewed findings to CSV, GeoJSON or a dated report. Inspection outcomes and resolution photos remain linked.

Example from upload to action

Illustrative example: a collector photographs a road surface. AI flags a possible pothole with confidence 0.87. GIS attaches the approximate photo GPS and suggests a nearby road segment and school context from imported layers. A reviewer confirms the pothole, corrects its pin and records severity under the agreed rubric. The project manager sees a verified finding with a transparent inspection priority, assigns a site visit and exports the record. All values in the UI concept are sample values.

Implementation responsibilities

Frontend: upload feedback, synchronized map/list, image overlay and review form. Backend: authorized storage, metadata, asynchronous jobs, access control and review history. ML engineer: label guide, geographic train/test separation, per-class model evaluation and versioned inference. GIS engineer: geographic layer ingestion, location quality, spatial associations, duplicate suggestions and geographic exports. QA: test missing GPS, wrong road matches, duplicate candidates, model failures, conflicting reviews and cross-organization access.

UI behavior and visual direction

The accompanying interactive concept uses a restrained green accent, clear left navigation and three key screens. Map and findings combines selectable observations with a source-image drawer. Review workspace separates the original evidence from editable human decisions. Photo collection shows location readiness and per-file progress. The concept uses schematic locations and illustrated imagery; it has no live AI model, geographic data service or upload backend. Verification, filtering and screen switching demonstrate intended behavior locally.

Technical references

PyTorch object detection tutorial: https://docs.pytorch.org/tutorials/intermediate/torchvision_tutorial.html

PostGIS geographic distance queries: https://postgis.net/docs/manual-3.5/en/ST_DWithin.html